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Inventor(s)

TAKEUCHI; Shunya et al.

CONNECTOR

Abstract

A connector is to be connected to a wire including a core wire and an insulation coating covering an outer periphery of the core wire and extending in a front-rear direction, and provided with a terminal to be connected to the core wire, a connector housing for accommodating the terminal inside and a rear holder to be attached to a rear part of the connector housing. The terminal includes a terminal body and a slide portion slidable in the front-rear direction while being externally fit to the terminal body. The terminal body includes a sandwiching portion for sandwiching the core wire. The slide portion includes a core wire insertion opening, through which the core wire is inserted from behind, and a pressing portion for pressing the sandwiching portion against the core wire inserted through the core wire insertion opening.

Inventors:	TAKEUCHI; Shunya (Mie, JP), HARA; Teruo (Mie, JP), MATSUI; Hajime (Mie, JP)
Applicant:	AUTONETWORKS TECHNOLOGIES, LTD. (Mie, JP); SUMITOMO WIRING SYSTEMS, LTD. (Mie, JP); SUMITOMO ELECTRIC INDUSTRIES, LTD. (Osaka, JP)
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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a connector.

BACKGROUND

[0002] Conventionally, a female terminal described in Japanese Patent Laid-Open Publication No. 2019-145208 (Patent Document 1 below) is known. The female terminal is provided with a terminal body including deformable upper and lower connection pieces extending in an extension direction and a slide portion movable in the extension direction with respect to the terminal body. By moving the slide portion with a core wire of a wire disposed between the upper and lower connection pieces, upper and lower contact portions provided in the slide portion press the upper and lower connection pieces against the core wire and the terminal body and the wire can be electrically connected. Further, a connector provided with a connector housing for accommodating the female terminal and a rear holder to be attached to a rear end part in the extension direction of the connector housing is described as a connector provided with this female terminal in Patent Document 1 below.

PRIOR ART DOCUMENT

Patent Document

[0003] Patent Document 1: JP 2019-145208 A

SUMMARY OF THE INVENTION

Problems to be Solved

[0004] The above connector is manufactured as follows. First, the female terminal is accommodated into the connector housing and the rear holder is assembled with the rear end part of the connector housing. In this state, the rear holder is held at a temporary locking position with respect to the connector housing. Subsequently, a wire is inserted into an insertion hole of the rear holder and a core wire of the wire is inserted into the inside of the female terminal. At this time, the core wire is disposed between the upper and lower connection pieces. Subsequently, by moving the slide portion forward with respect to the terminal body, the upper and lower contact portions press the upper and lower connection pieces against the core wire and the female terminal and the wire are connected. Finally, by moving the rear holder to a complete locking position forward of the temporary locking position to retain the female terminal in the connector housing, the manufacturing of the connector is completed.

[0005] In manufacturing the above connector, it is thought to provide a guide portion for guiding the insertion of the core wire into the inside of the female terminal on the inner wall of the insertion hole of the rear holder in order to reliably insert the core wire into the inside of the female terminal. For example, the guide portion is shaped to be located more inwardly of the insertion hole toward the front, and guides the core wire into the inside of the female terminal by sliding in contact with the core wire.

[0006] However, if an attempt is made to move the rear holder further rearward than the temporary locking position in the connector provided with the above guide portion, the guide portion may interfere with an insulation coating of the wire and it may not be possible to move the rear holder rearward. That is, it may not be possible to remove the rear holder from the connector. Thus, it

becomes difficult to perform, for example, a repair operation of the female terminal and the like.

Means to Solve the Problem

[0007] The present disclosure is directed to a connector to be connected to a wire including a core wire and an insulation coating covering an outer periphery of the core wire and extending in a front-rear direction, the connector being provided with a terminal to be connected to the core wire, a connector housing for accommodating the terminal inside and a rear holder to be attached to a rear part of the connector housing, the terminal including a terminal body and a slide portion slidable in the front-rear direction while being externally fit to the terminal body, the terminal body including a sandwiching portion for sandwiching the core wire, the slide portion including a core wire insertion opening, the core wire being inserted through the core wire insertion opening from behind, and a pressing portion for pressing the sandwiching portion against the core wire inserted through the core wire insertion opening, the rear holder including a body portion, an insertion hole penetrating through the body portion in the front-rear direction, the wire being inserted through the insertion hole, and a resilient piece extending from the body portion, the resilient piece being disposed on a front side inside the insertion hole, the resilient piece being resiliently deformable in a direction intersecting an extension direction of the wire, and the resilient piece being formed with a guide portion inclined inwardly of the insertion hole toward front.

Effect of the Invention

[0008] According to the present disclosure, it is possible to provide a connector capable of guiding the insertion of a core wire into the inside of a terminal accommodated in a connector housing and including an easily removable rear holder.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a perspective view of a connector according to an embodiment.

[0010] FIG. 2 is a perspective view showing a state where a rear holder is held on a connector housing at a temporary locking position.

[0011] FIG. 3 is a section showing a state where a terminal having a slide portion disposed at a separation position with respect to a terminal body is accommodated in the connector housing and the rear holder is held at the temporary locking position with respect to the connector housing.

[0012] FIG. 4 is a section showing a state where a core wire is inserted into an insertion hole from the state of FIG. 3 and a front end part of the core wire slides in contact with a guide portion.

[0013] FIG. 5 is a section showing a state where the core wire is inserted into the inside of the terminal from the state of FIG. 4.

[0014] FIG. 6 is a section showing a state where the slide portion is moved to a pressing position with respect to the terminal body from the state of FIG. 5.

[0015] FIG. 7 is a section showing a state where the rear holder is moved to a complete locking position with respect to the connector housing from the state of FIG. 6.

[0016] FIG. 8 is a section showing a state where the rear holder is moved further rearward than the temporary locking position with respect to the connector housing after the manufacturing of the connector.

[0017] FIG. 9 is a view of the terminal when viewed from behind the rear holder.

[0018] FIG. 10 is a back view of the rear holder.

[0019] FIG. 11 is a perspective view of the rear holder.

[0020] FIG. 12 is a section of a resilient piece.

[0021] FIG. 13 is a perspective view showing the resilient pieces with an upper part of the rear holder omitted.

[0022] FIG. 14 is a perspective view of the slide portion.

[0023] FIG. **15** is a perspective view of the terminal body.

[0024] FIG. **16** is a side view of the terminal having the slide portion disposed at the pressing position with respect to the terminal body.

[0025] FIG. **17** is a side view of the terminal having the slide portion disposed at the separation position with respect to the terminal body.

DETAILED DESCRIPTION TO EXECUTE THE INVENTION

DESCRIPTION OF EMBODIMENTS OF PRESENT DISCLOSURE

[0026] First, embodiments of the present disclosure are listed and described.

[0027] (1) The connector of the present disclosure is to be connected to a wire including a core wire and an insulation coating covering an outer periphery of the core wire and extending in a front-rear direction, and provided with a terminal to be connected to the core wire, a connector housing for accommodating the terminal inside and a rear holder to be attached to a rear part of the connector housing, the terminal including a terminal body and a slide portion slidable in the front-rear direction while being externally fit to the terminal body, the terminal body including a sandwiching portion for sandwiching the core wire, the slide portion including a core wire insertion opening, the core wire being inserted through the core wire insertion opening from behind, and a pressing portion for pressing the sandwiching portion against the core wire inserted through the core wire insertion opening, the rear holder including a body portion, an insertion hole penetrating through the body portion in the front-rear direction, the wire being inserted through the insertion hole, and a resilient piece extending from the body portion, the resilient piece being disposed on a front side inside the insertion hole, the resilient piece being resiliently deformable in a direction intersecting an extension direction of the wire, and the resilient piece being formed with a guide portion inclined inwardly of the insertion hole toward front.

[0028] According to this configuration, the insertion of the core wire into the core wire insertion opening can be guided by the guide portion. If the rear holder is provided with the guide portion, the guide portion contacts the insulation coating when the rear holder is separated from the connector housing for the repair of the terminal or the like. Since the guide portion is provided on the resilient piece resiliently deformable in the direction intersecting the extension direction of the wire in the above configuration, the rear holder can be separated from the connector housing by the insulation coating resiliently deforming the resilient piece.

[0029] (2) Preferably, a tip part of the guide portion is proximate to the core wire insertion opening.

[0030] According to this configuration, the insertion of the core wire into the core wire insertion opening can be more easily guided by the guide portion.

[0031] (3) Preferably, the guide portion has a curved surface shape.

[0032] According to this configuration, the core wire is easily guided toward a center of the core wire insertion opening.

[0033] (4) Preferably, a radius of curvature of the tip part of the guide portion is smaller than a radius of curvature of a rear end part of the guide portion.

[0034] According to this configuration, the core wire is even more easily guided toward the center of the core wire insertion opening by the tip part of the guide portion.

[0035] (5) Preferably, the guide portion includes a first guide portion and a second guide portion continuously disposed in front of the first guide portion, and the second guide portion is more gently inclined than the first guide portion.

[0036] According to this configuration, the inclination of the core wire with respect to the insertion direction can be suppressed by the second guide portion near the core wire insertion opening while the position of the core wire is guided inwardly of the core wire insertion opening by the first guide portion. Thus, the insertion of the core wire into the core wire insertion opening is even more easily guided.

[0037] (6) Preferably, the rear holder includes a guiding portion projecting inwardly of the insertion hole from an inner wall of the insertion hole toward the front, and the guiding portion is disposed to

face the guide portion in a resiliently deformable direction of the resilient piece.

[0038] According to this configuration, the insertion of the core wire into the core wire insertion opening can be even more easily guided by the guiding portion.

Details of Embodiment of Present Disclosure

[0039] Hereinafter, an embodiment of the present disclosure is described. The present disclosure is not limited to these illustrations, but is represented by claims and intended to include all changes in the scope of claims and in the meaning and scope of equivalents.

Embodiment

[0040] The embodiment of the present disclosure is described with reference to FIGS. **1** to **17**. In the following description, a direction indicated by an arrow **Z** is referred to as an upward direction, a direction indicated by an arrow **X** is referred to as a forward direction and a direction indicated by an arrow **Y** is referred to as a leftward direction. Note that, for a plurality of identical members, only some members may be denoted by a reference sign and the other members may not be denoted by the reference sign.

Connector

[0041] As shown in FIG. **1**, a connector **10** according to this embodiment is provided with terminals **12** connected to ends of wires **11**, a connector housing **30** including accommodating portions **32** for accommodating the terminals **12** and a rear holder **31** to be attached to a rear part of the connector housing **30**.

Wire

[0042] As shown in FIG. **5**, the wire **11** is disposed to extend in the front-rear direction. The wire **11** is configured such that the outer peripheral surface of the core wire **13** is surrounded by an insulation coating **14** made of insulating synthetic resin. The core wire **13** is made of electrically conductive metal and constituted by a stranded wire formed by twisting a plurality of metal wires or one metal wire. In a front end part of the wire **11**, the insulation coating **14** is removed to expose the core wire **13**.

Terminal

[0043] The terminal **12** is made of electrically conductive metal. As shown in FIGS. **16** and **17**, the terminal **12** includes the terminal body **15** and a slide portion **16** movable in the front-rear direction with respect to the terminal body **15**. The terminal body **15** and the slide portion **16** are formed into a predetermined shape by a known method such as press-working, cutting or casting. An arbitrary metal such as copper, copper alloy, aluminum, aluminum alloy or stainless steel can be appropriately selected as a metal constituting the terminal body **15** and the slide portion **16** if necessary. Plating layers may be formed on the surfaces of the terminal body **15** and the slide portion **16**. An arbitrary metal such as tin, nickel or silver can be appropriately selected as a metal constituting the plating layers if necessary.

Terminal Body

[0044] As shown in FIG. **15**, a front part of the terminal body **15** serves as a connecting tube portion **17** in the form of a rectangular tube extending in the front-rear direction. A plate-like mating terminal (not shown) can be inserted into the connecting tube portion **17** from front. An unillustrated resilient contact piece is disposed inside the connecting tube portion **17** and the mating terminal inserted into the connecting tube portion **17** contacts the resilient contact piece. A locking lance **21** having a chevron shape projecting upward is formed on a front side of the upper wall of the connecting tube portion **17**.

[0045] A base portion **20** in the form of a rectangular tube is provided behind the connecting tube portion **17**. A locking projection **28** projecting outward is formed on a side wall of the base portion **20**. A sandwiching portion **18** is provided in a rear end part of the base portion **20**.

Sandwiching Portion

[0046] The sandwiching portion **18** includes an upper connection piece **18A** extending rearward from a rear end part of the upper wall of the base portion **20** and a lower connection piece **18B**

extending rearward from a rear end part of the lower wall of the base portion **20**. The upper and lower connection pieces **18A**, **18B** have a shape elongated in the front-rear direction and have substantially equal lengths in the front-rear direction. The upper and lower connection pieces **18A**, **18B** are formed to be resiliently deformable in the vertical direction with a rear end part of the base portion **20** as a fulcrum. As shown in FIG. **6**, the lower surface of the upper connection piece **18A** and the upper surface of the lower connection piece **18B** can sandwich the core wire **13** and electrically connect the wire **11** and the terminal body **15**.

[0047] As shown in FIG. **15**, an upper holding protrusion **23A** projecting downward is provided at a position in front of a rear end part on the lower surface of the upper connection piece **18A**. A lower holding protrusion **23B** projecting upward is provided on a rear end part of the upper surface of the lower connection piece **18B**. The upper and lower holding protrusions **23A**, **23B** are provided at positions shifted in the front-rear direction.

Slide Portion

[0048] As shown in FIG. **14**, the slide portion **16** is in the form of a rectangular tube extending in the front-rear direction. As shown in FIGS. **3** and **6**, an internal cross-sectional shape of the slide portion **16** is the same as or somewhat larger than an external cross-sectional shape of a region of the terminal body **15** where the sandwiching portion **18** is provided. In this way, the slide portion **16** is fittable to the region of the terminal body **15** where the sandwiching portion **18** is provided.

Pressing Portion

[0049] As shown in FIG. **6**, the slide portion **16** is provided with a pressing portion **25**. In the pressing portion **25**, an upper pressing portion **25A** projecting downward is arranged on the lower surface of the upper wall of the slide portion **16**, and a lower pressing portion **25B** projecting upward is arranged on the upper surface of the lower wall of the slide portion **16**.

[0050] As shown in FIG. **14**, a temporary lock receiving portion **26** is open at a position near a front end part in a side wall of the slide portion **16**. Further, a complete lock receiving portion **27** is open at a position behind the temporary lock receiving portion **26** in the side wall of the slide portion **16**. The temporary lock receiving portion **26** and the complete lock receiving portion **27** are resiliently lockable by the locking projection **28** of the terminal body **15**.

[0051] As shown in FIG. **17**, with the locking projection **28** of the terminal body **15** and the temporary lock receiving portion **26** of the slide portion **16** locked, the slide portion **16** is held at a separation position with respect to the terminal body **15**. In this state, as shown in FIG. **5**, the pressing portion **25** of the slide portion **16** is disposed behind the rear end edge of the sandwiching portion **18** of the terminal body **15**. Further, in this state, an interval between the upper and lower connection pieces **18A**, **18B** is set to be larger than a diameter of the core wire **13**.

[0052] As shown in FIG. **16**, with the locking projection **28** of the terminal body **15** and the complete lock receiving portion **27** of the slide portion **16** locked, the slide portion **16** is held at a pressing position with respect to the terminal body **15**. In this state, as shown in FIG. **6**, the upper pressing portion **25A** of the slide portion **16** is in contact with the upper surface of the upper connection piece **18A** of the terminal body **15** from above. Further, the lower pressing portion **25B** of the slide portion **16** is in contact with the lower surface of the lower connection piece **18B** of the terminal body **15** from below.

[0053] With the slide portion **16** held at the pressing position with respect to the terminal body **15**, the upper pressing portion **25A** presses the upper connection piece **18A** from above, whereby the upper connection piece **18A** is resiliently deformed downward. Further, the lower pressing portion **25B** presses the lower connection piece **18B** from below, whereby the lower connection piece **18B** is resiliently deformed upward. In this way, the core wire **13** is disposed to extend in the front-rear direction in a space between the upper and lower connection pieces **18A**, **18B** and, with the slide portion **16** held at the pressing position with respect to the terminal body **15**, the core wire **13** is sandwiched in the vertical direction by the resiliently deformed upper and lower connection pieces **18A**, **18B**.

[0054] With the slide portion **16** held at the pressing position with respect to the terminal body **15**, the upper holding protrusion **23A** of the upper connection piece **18A** presses the core wire **13** from above and the lower holding protrusion **23B** of the lower connection piece **18B** presses the core wire **13** from below. As just described, the core wire **13** is held in a state bent in the vertical direction by being pressed from above by the upper holding protrusion **23A** and from below by the lower holding protrusion **23B** disposed at the position shifted in the front-rear direction from the upper holding protrusion **23A**. In this way, a holding force for holding the core wire **13** in the terminal body **15** can be improved.

[0055] As shown in FIG. **14**, a jig contact portion **24** projecting upward from the upper wall is provided in a front end part of the slide portion **16**. An unillustrated jig is brought into contact with the jig contact portion **24** from behind and the slide portion **16** is pushed forward by the jig, whereby the slide portion **16** becomes movable forward.

Core Wire Insertion Opening

[0056] A rear end part of the slide portion **16** in the form of a rectangular tube is formed with a core wire insertion opening **29**, through which the core wire **13** is inserted into the inside of the slide portion **16**.

Connector Housing

[0057] As shown in FIG. **1**, the connector housing **30** has a rectangular parallelepiped shape short in the vertical direction and long in the lateral direction. The connector housing **30** is made of insulating synthetic resin. The connector housing **30** includes a plurality of accommodating portions **32** extending in the front-rear direction for accommodating the terminals **12**. The accommodating portions **32** are aligned at intervals in the lateral direction in two upper and lower stages. The respective accommodating portions **32** formed in the upper stage and those formed in the lower stage are disposed at positions shifted in the lateral direction. Note that the number and relative arrangement of the accommodating portions **32** can be changed as appropriate.

[0058] As shown in FIG. **3**, the connector housing **30** is provided with lance receiving portions **33** corresponding to the positions of the locking lances **21** of the terminals **12** with the terminals **12** accommodated in the accommodating portions **32**. The locking lance **21** is locked by the lance receiving portion **33** and the terminal **12** is prevented from coming out rearward.

[0059] Although not shown, a front end part of the connector housing **30** is formed with front stop portions to be contacted by front end parts of the terminals **12**. When the terminal **12** is inserted into the accommodating portion **32**, the terminal **12** comes into contact with the front stop portion so as not come out forward from the connector housing **30**.

[0060] As shown in FIG. **3**, the connector housing **30** includes a separation wall **41** partitioning between the accommodating portions **32** formed in the upper stage and those formed in the lower stage. As shown in FIG. **1**, the connector housing **30** includes partition walls **42** partitioning between the respective accommodating portions **32** in the lateral direction. The terminals **12** accommodated into the respective accommodating portions **32** are electrically insulated from the terminals **12** adjacent in the lateral direction by the partition walls **42**.

[0061] As shown in FIGS. **1** and **2**, temporary lock portions **36** and complete lock portions **37** are provided to project outward on both left and right side walls of the connector housing **30**. The temporary lock portions **36** are disposed at positions near the rear end part of the connector housing **30**, and the complete lock portions **37** are disposed forward of the temporary lock portions **36**.

Rear Holder

[0062] As shown in FIG. **11**, the rear holder **31** is in the form of a box open forward. The rear holder **31** is made of insulating synthetic resin. As shown in FIG. **1**, the rear holder **31** is fit to a rear half of the connector housing **30** from outside. Lock receiving portions **38** are provided at positions near front end parts on both left and right walls of the rear holder **31**. The lock receiving portion **38** is substantially gate-shaped.

[0063] As shown in FIG. **2**, the rear holder **31** is held at a temporary locking position with respect

to the connector housing **30** by locking the temporary lock portions **36** of the connector housing **30** and the lock receiving portions **38** of the rear holder **31**. Further, as shown in FIG. **1**, the rear holder **31** is held at a complete locking position with respect to the connector housing **30** by locking the complete lock portions **37** of the connector housing **30** and the lock receiving portions **38** of the rear holder **31**.

[0064] As shown in FIG. **7**, the rear holder **31** is provided with a receptacle **43** open forward. The connector housing **30** is fit into the receptacle **43**. The rear holder **31** includes locking portions **44** in a rear end part of the receptacle **43**. The locking portions **44** are facing rear end parts of the terminals **12** in the front-rear direction. The locking portions **44** are locked to the rear end parts of the terminals **12** in the front-rear direction, whereby the terminals **12** are so retained in the accommodating portions **32** as not to come out rearward.

Body Portion, Insertion Holes

[0065] As shown in FIG. **10**, the rear holder **31** includes a body portion **45** and a plurality of insertion holes **46** penetrating through the body portion **45** in the front-rear direction. The insertion holes **46** are provided at positions corresponding to the respective accommodating portions **32** of the connector housing **30**. That is, the plurality of insertion holes **46** are aligned in the lateral direction in two upper and lower stages. An inner diameter of the insertion hole **46** is set to be equal to or somewhat larger than an outer diameter of the insulation coating **14** of the wire **11**. As shown in FIGS. **4** to **8**, the wire **11** is inserted into the insertion hole **46**. The insertion holes **46** are disposed to communicate with the respective accommodating portions **32** of the connector housing **30** in the front-rear direction.

Resilient Pieces

[0066] As shown in FIG. **12**, the rear holder **31** includes a plurality of resilient pieces **47** extending forward from a vertically central part of the body portion **45**. Each resilient piece **47** is disposed on a front side inside each insertion hole **46**. As shown in FIG. **10**, the resilient piece **47** is provided on a lower side in the insertion hole **46** in the upper stage, and the resilient piece **47** is provided on an upper side in the insertion hole **46** in the lower stage.

[0067] As shown in FIG. **12**, the resilient piece **47** is configured to be resiliently deformable in the vertical direction with a rear end part connected to the body portion **45** substantially as a fulcrum. An entrance space **48** is provided between the resilient pieces **47** in the upper and lower stages. Although described in detail later, the resilient piece **47** is resiliently deformed outwardly of the insertion hole **46** by being pressed by the insulation coating **14** of the wire **11** as shown in FIG. **8** in the case of separating the rear holder **31** from the connector housing **30** after the connector **10** is manufactured. That is, the resilient piece **47** in the upper stage is resiliently deformed downward. The resilient piece **47** in the lower stage is not resiliently deformed in FIG. **8**, but is resiliently deformed upward. The resilient piece **47** is disposed in the entrance space **48** if being resiliently deformed by being contacted by the insulation coating **14**. Note that a cross-section of the terminal **12** is not shown, but only the outer shape of the terminal **12** is shown in FIG. **8**.

[0068] As shown in FIG. **6**, with the rear holder **31** held at the temporary locking position with respect to the connector housing **30**, the rear end edge of the separation wall **41** of the connector housing **30** is disposed in front of the entrance space **48**. However, since gaps are present between the separation wall **41** and the resilient pieces **47**, the resilient deformation of the resilient pieces **47** is not hindered by the separation wall **41**. As shown in FIG. **7**, with the rear holder **31** held at the complete locking position with respect to the connector housing **30**, the separation wall **41** is inserted to the back of the entrance space **48**.

[0069] Since the resilient pieces **47** in the upper stage and those in the lower stage are symmetrically configured, the configuration of the resilient piece **47** is described based on a positional relationship in the upper stage unless otherwise stated.

Guide Portion

[0070] As shown in FIG. **12**, a guide portion **49** is formed on the upper surface of the resilient piece

47. The guide portion 49 is inclined inwardly (here, upwardly) of the insertion hole 46 toward the front. As shown in FIG. 3, the terminal 12 having the slide portion 16 held at the separation position is accommodated into the accommodating portion 32 of the connector housing 30 and, with the rear holder 31 assembled at the temporary locking position with respect to the connector housing 30, a tip part of the guide portion 49 is disposed to face the core wire insertion opening 29 in the front-rear direction. Further, a front end part of the guide portion 49 is disposed in proximity to the core wire insertion opening 29. Therefore, if the wire 11 is inserted into the insertion hole 46, a front end part of the core wire 13 is guided into the core wire insertion opening 29 by sliding in contact with the guide portion 49 as shown in FIG. 4.

First Guide Portion, Second Guide Portion

[0071] As shown in FIG. 12, the guide portion 49 of this embodiment includes a first guide portion 50 and a second guide portion 51 formed continuously in front of the first guide portion 50. The second guide portion 51 is substantially one-fourth of the guide portion 49 on a front side. The first guide portion 50 is a part of the guide portion 49 except the second guide portion 51. An angle of inclination of the first guide portion 50 with respect to an insertion direction (front-rear direction), in which the wire 11 is inserted, is larger than that of the second guide portion 51. Thus, the position of the front end part of the core wire 13 can be largely moved inwardly of the core wire insertion opening 29 by the core wire 13 sliding in contact with the first guide portion 50. On the other hand, since the second guide portion 51 is set to have a small angle of inclination, the second guide portion 51 guides the core wire 13 in a posture extending substantially in the insertion direction into the core wire insertion opening 29 while finely adjusting the position of the front end part of the core wire 13 (see FIG. 4).

[0072] For example, if the core wire 13 faces the core wire insertion opening 29 while being largely inclined with respect to the insertion direction, the core wire 13 may not possibly enter the core wire insertion opening 29 by the front end part of the core wire 13 being caught by a rear end part of the upper wall of the slide portion 16. However, since the core wire 13 is caused to face the core wire insertion opening 29 in a state not largely inclined with respect to the insertion direction by the second guide portion 51 in this embodiment, the core wire 13 easily smoothly enters the core wire insertion opening 29.

Guiding Portion

[0073] As shown in FIG. 12, the rear holder 31 includes guiding portions 52 projecting inwardly of the insertion holes 46 from the inner walls of the insertion holes 46. Note that a broken line above the guiding portion 52 represents the inner wall of the insertion hole 46. The guiding portion 52 is disposed on a front side of the insertion hole 46 and located more inwardly (here, downwardly) of the insertion hole 46 toward the front. The core wire 13 easily enters the core wire insertion opening 29 by sliding in contact with the guiding portion 52. Further, the guiding portion 52 is facing the resilient piece 47 in the vertical direction. Thus, the core wire 13 can be guided into the core wire insertion opening 29 from a side opposite to the resilient piece 47 by the guiding portion 52.

[0074] As shown in FIG. 13, the guide portion 49 of this embodiment is in the form of a curved surface concave downward. The guide portion 49 is in the form of a groove extending in the front-rear direction and a substantially U-shaped cross-section. According to this shape, the core wire 13 is easily guided toward a center in a width direction (lateral direction) of the guide portion 49 by the guide portion 49. That is, the core wire 13 is easily guided toward a center of the core wire insertion opening 29.

[0075] Further, in this embodiment, a radius of curvature of the tip part of the guide portion 49 is smaller than that of the rear end part of the guide portion 49 as shown in FIG. 9. Therefore, the tip part of the guide portion 49 more easily guides the core wire 13 toward the core wire insertion opening 29.

Connector Manufacturing Method

[0076] Next, an example of a manufacturing method of the connector **10** according to this embodiment is described.

[0077] The terminal body **15** and the slide portion **16** are formed by a known method. The slide portion **16** is assembled with the terminal body **15** from behind. The front end edge of the slide portion **16** comes into contact with the locking projection **28** of the terminal body **15** from behind and the side wall of the slide portion **16** is expanded and deformed. If the slide portion **16** is pushed further forward, the side wall of the slide portion **16** is restored and the temporary lock receiving portion **26** of the slide portion **16** is locked to the locking projection **28** of the terminal body **15**. In this way, the slide portion **16** is held at the separation position with respect to the terminal body **15**. In this way, the terminal **12** is obtained (see FIG. 7).

[0078] By injection-molding the synthetic resin, the connector housing **30** and the rear holder **31** are formed. After the terminals **12** are inserted into the accommodating portions **32** of the connector housing **30** from behind, the rear holder **31** is assembled with the rear end part of the connector housing **30** from behind. Then, the lock receiving portions **38** of the rear holder **31** ride on the temporary lock portions **36** of the connector housing **30** while being resiliently deformed. If the rear holder **31** is pushed further forward, the lock receiving portions **38** are restored and resiliently locked to the temporary lock portions **36** of the connector housing **30**. In this way, the rear holder **31** is held at the temporary locking position with respect to the connector housing **30** (see FIGS. 2 and 3).

[0079] The core wire **13** is exposed over a predetermined length in the end part of the wire **11**. The front end part of the core wire **13** is inserted from behind into the insertion hole **46** disposed behind the accommodating portion **32** for accommodating the terminal **12**. As shown in FIG. 4, the core wire **13** inserted into the insertion hole **46** slides in contact with the guide portion **49** of the rear holder **31**. Since the guide portion **49** is inclined inwardly of the insertion hole **46** toward the front, the core wire **13** is guided into the core wire insertion opening **29**. The core wire **13** is guided into the core wire insertion opening **29** similarly by the guiding portion **52** provided to face the guide portion **49** in the vertical direction.

[0080] Since the guide portion **49** has a substantially U-shaped cross-section and is in the form of a groove, the core wire **13** is guided toward the center also in the width direction (lateral direction) of the resilient piece **47**. Thus, the core wire **13** is easily guided toward the center of the core wire insertion opening **29**.

[0081] Since the second guide portion **51** having a small angle of inclination with respect to the insertion hole of the wire **11** is provided on a tip side of the guide portion **49** in this embodiment, the core wire **13** approaching the core wire insertion opening **29** is in a posture not inclined very much with respect to the insertion direction of the wire **11**. In this way, the core wire **13** is more easily guided into the core wire insertion opening **29**. Further, since the radius of curvature of the guide portion **49** is smaller in the tip part of the guide portion **49** than in the rear end part of the guide portion **49**, the core wire **13** is even more easily guided toward the widthwise center of the guide portion **49** near the core wire insertion opening **29**.

[0082] As described above, the core wire **13** of the wire **11** inserted into the insertion hole **46** enters the core wire insertion opening **29** by sliding in contact with the guide portion **49**.

[0083] If the wire **11** is pushed further forward, the core wire **13** enters the inside of the terminal body **15** and is disposed at the position of the sandwiching portion **18** in the front-rear direction. That is, the core wire **13** passes between the upper and lower connection pieces **18A**, **18B** (see FIG. 5). Note that, although the front end part of the insulation coating **14** is located near the rear end part of the resilient piece **47** in FIG. 5, the insulation coating **14** may, for example, reach the front end part of the insertion hole **46** and deflect the resilient piece **47**.

[0084] Subsequently, the unillustrated jig is pushed into contact with the jig contact portion **24** from behind and the slide portion **16** is moved forward with respect to the terminal body **15**. At this time, the locking of the locking projection **28** of the terminal body **15** and the temporary lock

receiving portion **26** of the slide portion **16** is released and the side wall of the slide portion **16** rides on the locking projection **28** and is expanded and deformed.

[0085] If the jig contact portion **24** is pushed further forward by the jig, the side wall of the slide portion **16** is restored and the locking projection **28** of the terminal body **15** and the complete lock receiving portion **27** of the slide portion **16** are resiliently locked. In this way, the slide portion **16** is held at the pressing position with respect to the terminal body **15** (see FIG. 6).

[0086] When the slide portion **16** moves to the pressing position with respect to the terminal body **15**, the upper pressing portion **25A** of the slide portion **16** presses the upper connection piece **18A** of the terminal body **15** downward. Further, the lower pressing portion **25B** of the slide portion **16** presses the lower connection piece **18B** of the terminal body **15** upward. In this way, the core wire **13** is vertically sandwiched by the upper and lower connection pieces **18A**, **18B** and the wire **11** and the terminal **12** are electrically connected.

[0087] Subsequently, if the rear holder **31** is pushed forward, the lock receiving portions **38** of the rear holder **31** ride on the complete lock portions **37** of the connector housing **30** and are resiliently deformed. If the rear holder **31** is pushed further forward, the lock receiving portions **38** are locked to the complete lock portions **37**. In this way, the rear holder **31** is held at the complete locking position with respect to the connector housing **30** (see FIGS. 7 and 1). In the above way, the manufacturing of the connector **10** is completed.

[0088] In a state where the manufacturing of the connector **10** is completed, the resilient piece **47** is preferably in a natural state without being pressed by the insulation coating **14** as shown in FIG. 7. This is because the strength of the resilient piece **47** may be reduced with the resilient piece **47** resiliently deformed.

Repair Operation of Terminal

[0089] Next, an example of a repair operation of the terminal **12** in the connector **10** of this embodiment is described. Note that the repair of the terminal **12** means the exchange of the terminal **12** having a problem in the connector **10** for a new terminal **12**.

[0090] First, the rear holder **31** is removed from the connector housing **30**. The locking of the lock receiving portions **38** of the rear holder **31** with the complete lock portions **37** and the temporary lock portions **36** is released by deflecting the lock receiving portions **38** outward, and the rear holder **31** is moved rearward.

[0091] If the rear holder **31** is moved rearward, the guide portions **49** contact the insulation coatings **14** of the wires **11**. In this embodiment, since the guide portion **49** is provided on the resiliently deformable resilient piece **47**, the rear holder **31** can be moved further rearward and separated from the connector housing **30** by the insulation coatings **14** of the wires **11** resiliently deforming the resilient pieces **47** as shown in FIG. 8.

[0092] With the rear holder **31** separated rearward from the temporary locking position, the terminal **12** to be repaired can be taken out from the accommodating portion **32** since the space behind the accommodating portion **32** can be utilized. In particular, the locking lance **21** of the terminal body **15** is squashed with a jig or the like and the terminal **12** is pulled out rearward.

[0093] Subsequently, the new terminal **12** is accommodated into the accommodating portion **32**, and the connector **10** is configured again in the same way as the manufacturing method described above. Alternatively, the connector **10** may be configured by accommodating the new terminal **12** connected to the wire **11** in advance into the accommodating portion **32**.

Functions and Effects of Embodiment

[0094] According to the embodiment, the following functions and effects are achieved.

[0095] The connector **10** according to the embodiment is to be connected to the wires **11** each including the core wire **13** and the insulation coating **14** covering the outer periphery of the core wire **13** and extending in the front-rear direction, and provided with the terminals **12** to be connected to the core wires **13**, the connector housing **30** for accommodating the terminals **12** inside and the rear holder **31** to be attached to the rear part of the connector housing **30**. The

terminal **12** includes the terminal body **15** and the slide portion **16** slidable in the front-rear direction while being externally fit to the terminal body **15**. The terminal body **15** includes the sandwiching portion **18** for sandwiching the core wire **13**. The slide portion **16** includes the core wire insertion opening **29**, through which the core wire **13** is inserted from behind, and the pressing portion **25** for pressing the sandwiching portion **18** against the core wire **13** inserted through the core wire insertion opening **29**. The rear holder **31** includes the body portion **45**, the insertion holes **46** penetrating through the body portion **45** in the front-rear direction, the wires **11** being inserted inside the insertion holes **46**, and the resilient pieces **47** extending from the body portion **45** and disposed on the front sides inside the insertion holes **46**. The resilient piece **47** is resiliently deformable in a direction intersecting an extension direction of the wire **11** and formed with the guide portion **49** inclined inwardly of the insertion hole **46** toward the front.

[0096] According to this configuration, the insertion of the core wire **13** into the core wire insertion opening **29** can be guided by the guide portion **49**. If the rear holder **31** is provided with the guide portion **49**, the guide portion **49** contacts the insulation coating **14** when the rear holder **31** is separated from the connector housing **30** for the repair of the terminal **12** or the like. In the above configuration, since the guide portion **49** is provided on the resilient piece **47** resiliently deformable in the direction intersecting the extension direction of the wire **11**, the rear holder **31** is easily separated from the connector housing **30** by the insulation coating **14** resiliently deforming the resilient piece **47**.

[0097] In the embodiment, the tip part of the guide portion **49** is proximate to the core wire insertion opening **29**.

[0098] According to this configuration, the insertion of the core wire **13** into the core wire insertion opening **29** is more easily guided by the guide portion **49**.

[0099] In the embodiment, the guide portion **49** has a curved surface shape.

[0100] According to this configuration, the core wire **13** is easily guided toward the center of the core wire insertion opening **29**.

[0101] In the embodiment, the radius of curvature of the tip part of the guide portion **49** is smaller than that of the rear end part of the guide portion **49**.

[0102] According to this configuration, the core wire **13** is even more easily guided toward the center of the core wire insertion opening **29** by the tip part of the guide portion **49**.

[0103] In the embodiment, the guide portion **49** includes the first guide portion **50** and the second guide portion **51** disposed continuously in front of the first guide portion **50**, and the second guide portion **51** is more gently inclined than the first guide portion **50**.

[0104] According to this configuration, the inclination of the core wire **13** with respect to the insertion direction can be suppressed by the second guide portion **51** near the core wire insertion opening **29** while the position of the core wire **13** is guided inwardly of the core wire insertion opening **29** by the first guide portion **50**. Thus, the insertion of the core wire **13** into the core wire insertion opening **29** is even more easily guided.

[0105] In the embodiment, the rear holder **31** includes the guiding portions **52** projecting inwardly of the insertion holes **46** from the inner walls of the insertion holes **46** toward the front, and the guiding portion **52** is disposed to face the guide portion **49** in a resiliently deformable direction of the resilient piece **47**.

[0106] According to this configuration, the insertion of the core wire **13** into the core wire insertion opening **29** is even more easily guided by the guiding portion **52**.

Other Embodiments

[0107] (1) Although one resilient piece **47** is provided in each insertion hole **46** and disposed on the upper or lower side of each insertion hole **46** in the above embodiment, there is no limitation to this and the number and positions of the resilient pieces provided in each insertion hole may be changed as appropriate. For example, two resilient pieces may be provided on the upper and lower sides of each insertion hole. [0108] (2) Although the rear holder **31** is held at the temporary locking position

and the complete locking position with respect to the connector housing **30** in the above embodiment, there is no limitation to this and a rear holder may be held at one locking position with respect to the connector housing.

LIST OF REFERENCE NUMERALS

[0109] **10**: connector [0110] **11**: wire [0111] **12**: terminal [0112] **13**: core wire [0113] **14**: insulation coating [0114] **15**: terminal body [0115] **16**: slide portion [0116] **17**: connecting tube portion [0117] **18**: sandwiching portion [0118] **18A**: upper connection piece [0119] **18B**: lower connection piece [0120] **20**: base portion [0121] **21**: locking lance [0122] **23A**: upper holding protrusion [0123] **23B**: lower holding protrusion [0124] **24**: jig contact portion [0125] **25**: pressing portion [0126] **25A**: upper pressing portion [0127] **25B**: lower pressing portion [0128] **26**: temporary lock receiving portion [0129] **27**: complete lock receiving portion [0130] **28**: locking projection [0131] **29** core wire insertion opening [0132] **30**: connector housing [0133] **31**: rear holder [0134] **32**: accommodating portion [0135] **33**: lance receiving portion [0136] **36**: temporary lock portion [0137] **37**: complete lock portion [0138] **38**: lock receiving portion [0139] **41** separation wall [0140] **42** partition wall [0141] **43**: receptacle [0142] **44**: locking portion [0143] **45**: body portion [0144] **46** insertion hole [0145] **47** resilient piece [0146] **48**: entrance space [0147] **49**: guide portion [0148] **50**: first guide portion [0149] **51**: second guide portion [0150] **52**: guiding portion

Claims

1. A connector to be connected to a wire including a core wire and an insulation coating covering an outer periphery of the core wire and extending in a front-rear direction, comprising: a terminal to be connected to the core wire; a connector housing for accommodating the terminal inside; and a rear holder to be attached to a rear part of the connector housing, the terminal including a terminal body and a slide portion slidable in the front-rear direction while being externally fit to the terminal body, the terminal body including a sandwiching portion for sandwiching the core wire, the slide portion including a core wire insertion opening, the core wire being inserted through the core wire insertion opening from behind, and a pressing portion for pressing the sandwiching portion against the core wire inserted through the core wire insertion opening, the rear holder including a body portion, an insertion hole penetrating through the body portion in the front-rear direction, the wire being inserted through the insertion hole, and a resilient piece extending from the body portion, the resilient piece being disposed on a front side inside the insertion hole, the resilient piece being resiliently deformable in a direction intersecting an extension direction of the wire, and the resilient piece being formed with a guide portion inclined inwardly of the insertion hole toward front.
 2. The connector of claim 1, wherein a tip part of the guide portion is proximate to the core wire insertion opening.
 3. The connector of claim 1, wherein the guide portion has a curved surface shape.
 4. The connector of claim 3, wherein a radius of curvature of a tip part of the guide portion is smaller than a radius of curvature of a rear end part of the guide portion.
 5. The connector of claim 1, wherein: the guide portion includes a first guide portion and a second guide portion continuously disposed in front of the first guide portion, and the second guide portion is more gently inclined than the first guide portion.
 6. The connector of claim 1, wherein: the rear holder includes a guiding portion projecting inwardly of the insertion hole from an inner wall of the insertion hole toward the front, and the guiding portion is disposed to face the guide portion in a resiliently deformable direction of the resilient piece.
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